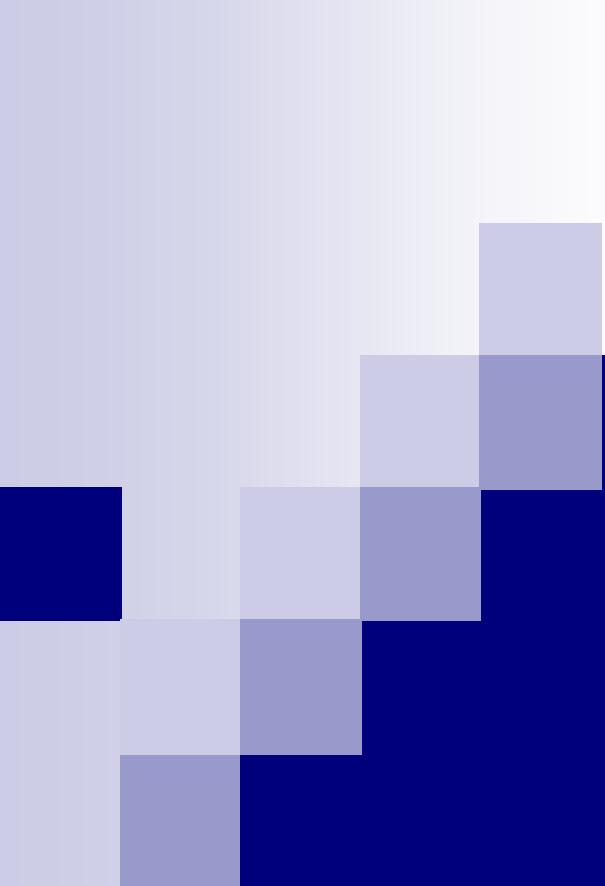
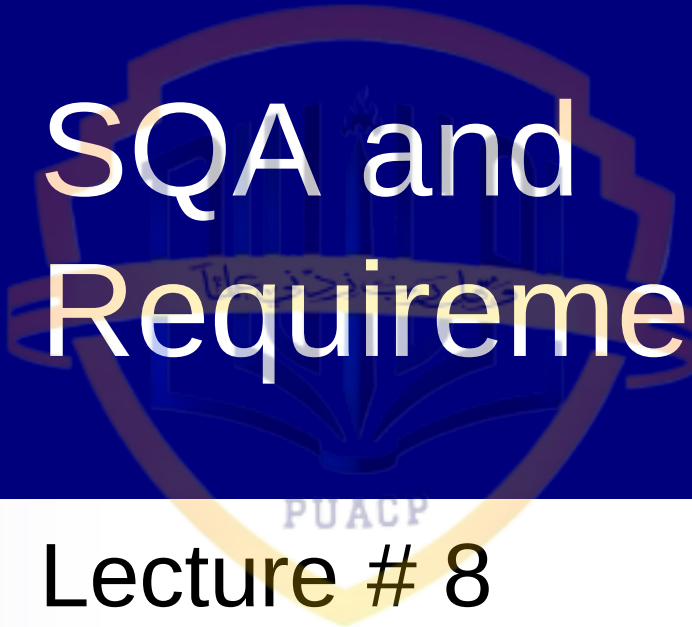




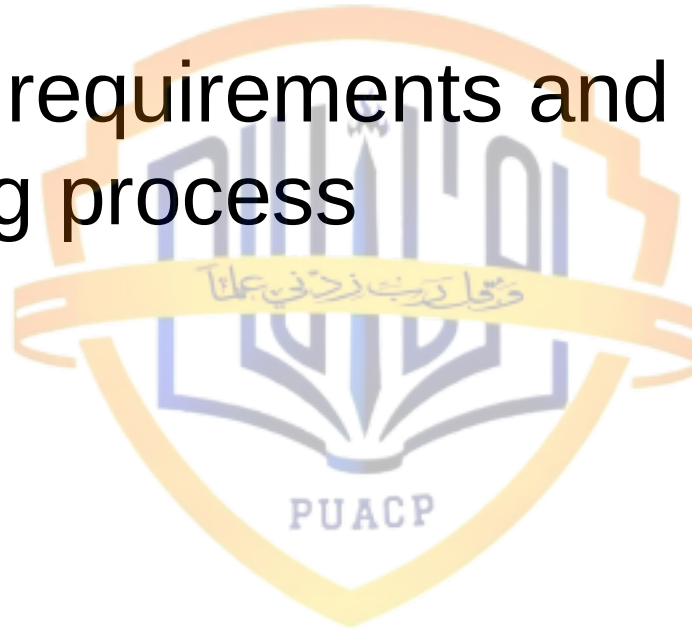
SQA and Requirements



Lecture # 8

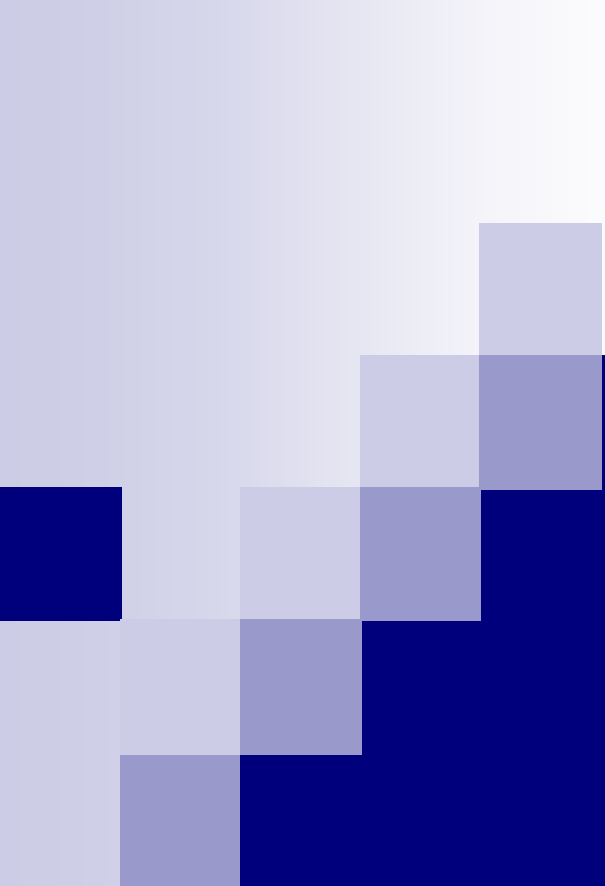
Recap

- Discussed requirements and requirements engineering process



Today's Lecture

- Problems in requirements and requirements defects
- Some guidelines for writing requirements to improve the quality of the requirements document



Requirements Problems and Requirements Defects



Requirements Problems - 1

- The requirements don't reflect the real needs of the customer for the system
- Requirements are inconsistent and/or incomplete
- It is expensive to make changes to requirements after they have been agreed upon

Requirements Problems - 2

- There are misunderstandings between customers, those developing the system requirements, and software engineers developing or maintaining the system
- The requirements are written using complex conditional clauses (if A then B then C...), which are confusing

Requirements Problems - 3

- Terminology is used in a sloppy and inconsistent way
- The writers of the requirement assume that the reader has a specific knowledge of the domain or the system and they leave essential information out of the requirements document

Impact of Wrong Requirements - 1

- Difficult to check the requirements for errors and omissions
- Different interpretations of the requirements may lead to contractual disagreements between customer and the system developer

Impact of Wrong Requirements - 2

- When requirements are wrong, systems are late, unreliable and don't meet customers needs
- This results in enormous loss of time, revenue, market share, and trust of customers
- Conformance to wrong requirements will result in a wrong system

Requirements Defects - 1

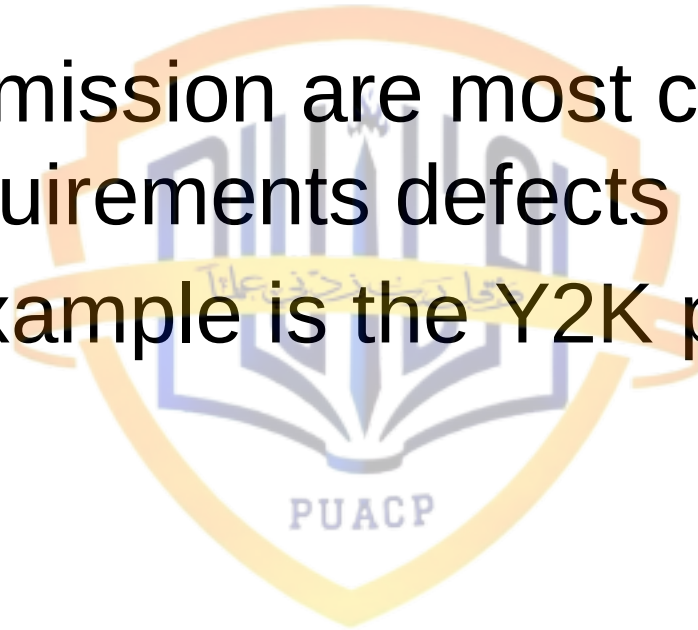
- All four categories of defects are found in requirements
 - Errors of commission
 - Errors of omission
 - Errors of clarity and ambiguity
 - Errors of speed and capacity
- If not prevented or removed, requirements defects usually flow downstream into design, code, and user manuals

Requirements Defects - 2

- Historically, requirements defects are most expensive and troublesome to eliminate
- Every effort should be made to eliminate requirements defects
- For requirements errors, prevention usually works better than removal

Requirements Defects - 3

- Errors of omission are most common among requirements defects
- Famous example is the Y2K problem



Requirements Defects - 4

- Second most common errors are those of clarity and ambiguity
- Primarily, because natural languages (like English) are used to state requirements, while such languages are themselves ambiguous
- For example: object

Requirements Defects - 5

- Errors of commission can also find their way into the requirements documents
- Performance errors can also be found in requirements documents

Prevention Vs Removal

- For requirements errors, prevention is usually more effective than removal
- Joint application development (JAD), Quality Function Deployment (QFD), and prototyping are more effective in defect prevention
- Requirements inspections and prototyping play an important role defect removal

Changing/Creeping Requirements - 1

- Requirements will change, no matter what
- A major issue in requirements engineering is the rate at which requirements change once the requirements phase has “officially” ended
- This rate is on average 3% per month in the subsequent design phase, and will go down after that

Changing/Creeping Requirements - 2

- This rate should come down to 1% per month during coding
- Ideally, this should come down to no changes in testing

Defects and Creeping Requirements

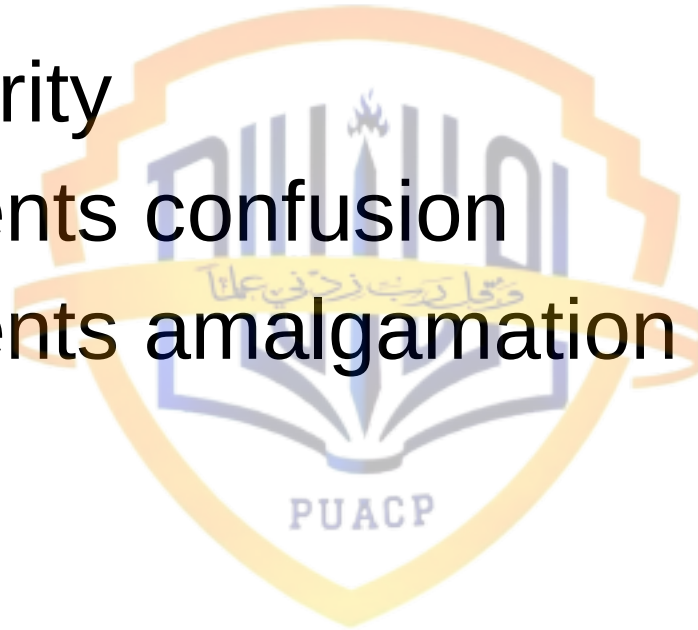
- Studies have shown that very significant percentage of delivered defects can be traced back to creeping user requirements
- This realization can only be made, if defect tracking, requirements traceability, defect removal efficiency, and defect rates are all monitored for software projects

Damage Control of Creeping Requirements

- Following quality assurance mechanisms can limit the damage done by creeping requirements
 - Formal change management procedures
 - State-of-the-art configuration control tools
 - Formal design and code inspections

Problems with Natural Languages - 1

- Lack of clarity
- Requirements confusion
- Requirements amalgamation

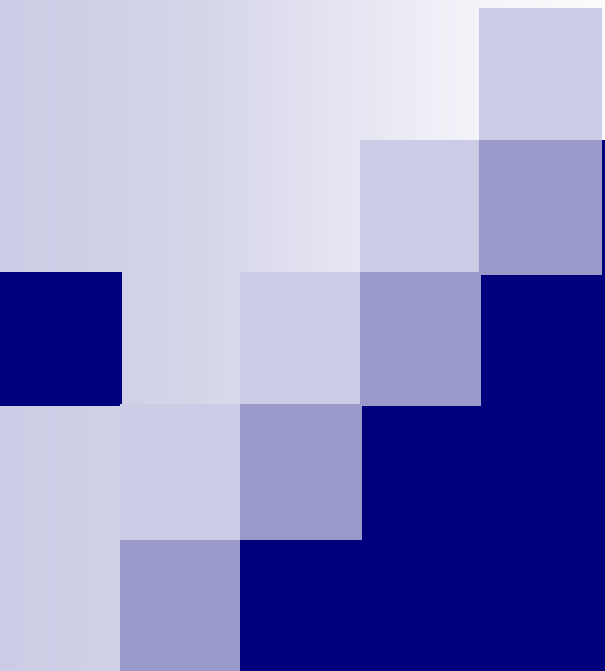


Problems with Natural Languages - 2

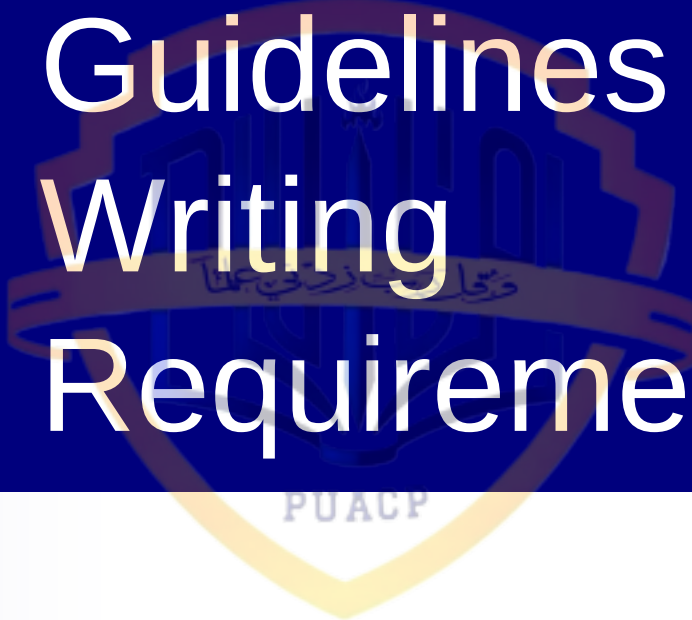
- Natural language understanding relies on the specification readers and writers using the same words for same concept
- A natural language requirements specification is over-flexible. You can say the same thing in completely different ways

Problems with Natural Languages - 3

- It is not possible to modularize natural language requirements. It may be difficult to find all related requirements
 - To discover the impact of a change, every requirement have to be examined



Guidelines for Writing Requirements



Writing Requirements - 1

- Requirements specification should establish an understanding between customers and suppliers about what a system is supposed to do, and provide a basis for validation and verification

Writing Requirements - 2

- Typically, requirements documents are written in natural languages (like, English, Japanese, French, etc.)
- Natural languages are ambiguous
- Structured languages can be used with the natural languages to specify requirements
 - These languages cannot completely define requirements
 - They are not understandable by all stakeholders

Essentials for Writing Requirements - 1

- Requirements are read more often than they are written. Investing effort in writing requirements, which are easy to read and understand is almost always cost-effective
- Readers of requirements come from diverse backgrounds. Requirements writers should not assume that readers have the same background and knowledge as them

Essentials for Writing Requirements - 2

- Writing clearly and concisely is not easy. If you don't allow sufficient time for requirements descriptions to be drafted, reviewed and improved, you will inevitably end up with poorly written requirements

Essentials for Writing Requirements - 3

- Different organizations write requirements at different levels of abstraction from deliberately vague product specifications to detailed and precise descriptions of all aspects of a system

Essentials for Writing Requirements - 4

- Level of detail needed is dependent on
 - Type of requirements (stakeholder or process requirements)
 - Customer expectations
 - Organizational procedures
 - External standards or regulations

Essentials for Writing Requirements - 5

- Writing good requirements requires a lot of analytic thought
- Specifying rationale of requirement is one way to encourage such thought

Guidelines for Writing Requirements - 1

- Define standard templates for describing requirements
- Use language simply, consistently, and concisely
- Use diagrams appropriately

Use of Standard Templates

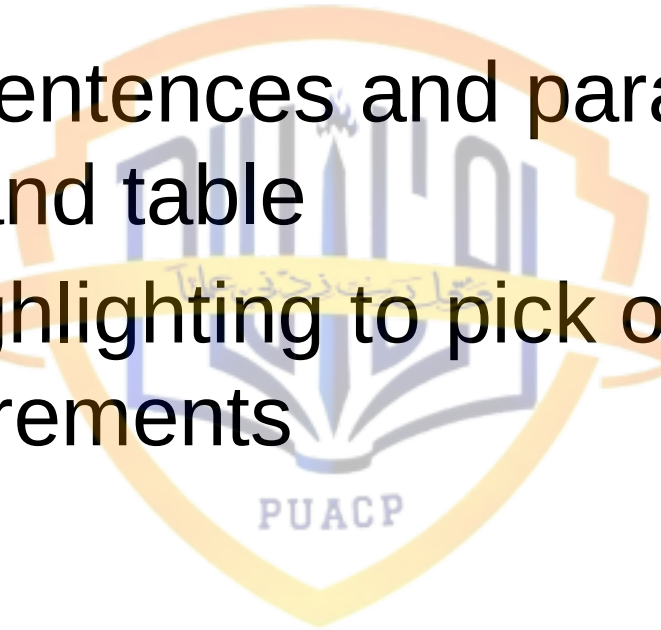
- Define a set of standard format for different types of requirements and ensure that all requirement definitions adhere to that format
- Standardization means that omissions are less likely and makes requirements easier to read and check

Using Simple Language - 1

- Use language consistently. In particular, distinguish between mandatory and desirable requirements. It is usual practice to define mandatory requirements using '*shall*' and desirable requirements using '*should*'. Use '*will*' to state facts or declare purpose

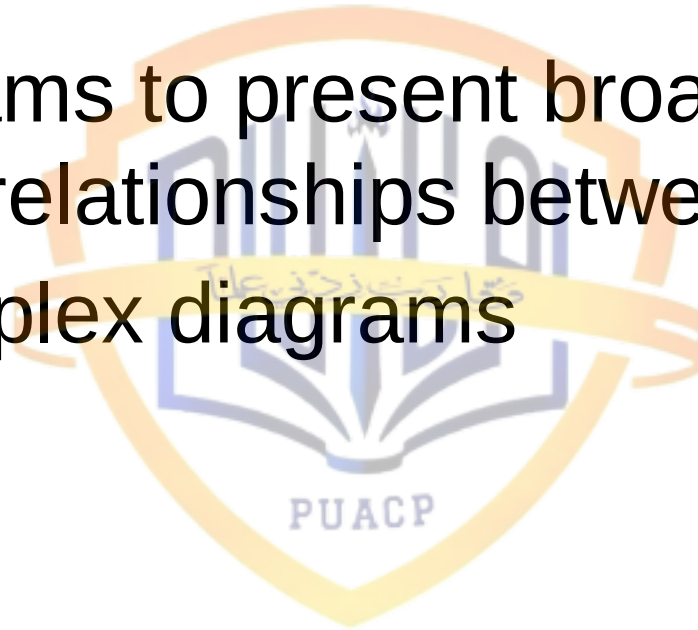
Using Simple Language - 2

- Use short sentences and paragraphs, using lists and table
- Use text highlighting to pick out key parts of the requirements



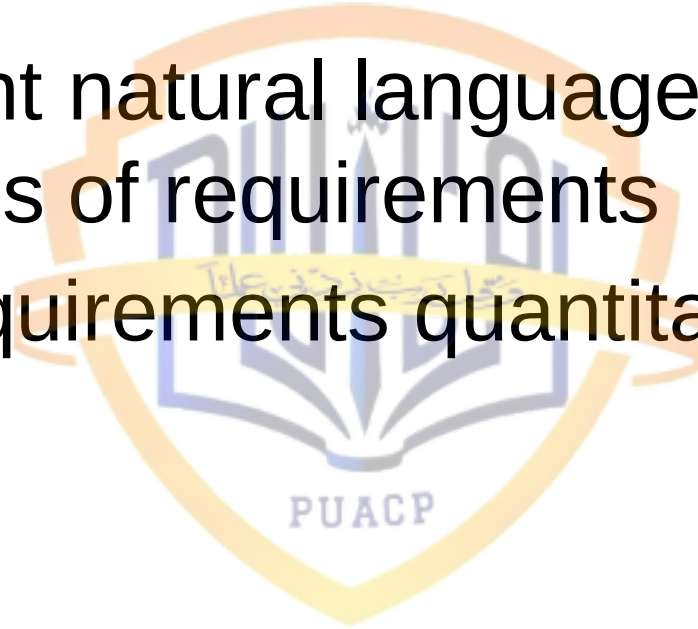
Using Appropriate Diagrams

- Use diagrams to present broad overviews and show relationships between entities
- Avoid complex diagrams



Guidelines for Writing Requirements - 2

- Supplement natural language with other descriptions of requirements
- Specify requirements quantitatively



Using Other Descriptions of Requirements

- If readers are familiar with other types of descriptions of requirements (like equations, etc.) then use those
- Particularly applicable to scientific and engineering domains
- Don't try to write everything in natural language

Specify Requirements Quantitatively

- Specify requirements quantitatively wherever possible
- This is applicable to properties of system, such as reliability or performance
- Recollect our discussion on metrics for non-functional requirements

Additional Guidelines for Writing Requirements - 1

- State only requirement per requirement statement
- State requirements as active sentences
- Always use a noun or a definite pronoun when referring to a thing
- Do not use more than one conjunction when writing requirements statements

Additional Guidelines for Writing Requirements - 2

- Avoid using weak words and phrases. Such words and phrases are generally imprecise and allow the expansion or contraction of requirements beyond their intent

Examples of Words to be Avoided

- About, adequate, and/or, appropriate, as applicable, as appropriate, desirable, efficient, etc., if practical, suitable, timely, typical, when necessary

Additional Guidelines for Writing Requirements - 3

- State the needed requirements without specifying how to fulfill them
- Write complete statements
- Write statements that clearly convey intent

References

- 'Requirements Engineering: Processes and Techniques' by G. Kotonya and I. Sommerville, John Wiley & Sons, 1998
- 'Software Requirements: Objects, Functions, and States' by A. Davis, PH, 1993
- 'Software Engineering Quality Practices' by R. K. Kandt, Auerbach Publications, 2006